Method	Spatial SNR ↑	SAM ↓	Pearson Correlation ↑	Processing Strategy	Generalization
Physics-Based Correction	25.4	0.125	0.72	Physics-based	CRISM only
HSID-CNN	31.2	0.084	0.84	Supervised Deep Learning	CRISM only
N2N4M	33.8	0.076	0.87	Self-supervised Deep Learning	CRISM only
Proposed Hybrid Framework	**52.6**	**0.042**	**0.96**	**Hybrid Physics + Self-supervised Learning**	**CRISM + M³**

Higher Spatial SNR and Pearson Correlation indicate better spatial restoration and spectral consistency, whereas a lower SAM indicates improved preservation of spectral signatures.